Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: P&D Dairy North

Physical address of dairy:

28366 Husman RD	Gustine	Merced	95322
Number and Street	City	County	Zip Code

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1945

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X070-X030-X020-XXXX

B. OPERATORS

Lopes, Paul

Operator name: Lopes, Paul

Telephone no.: Landline: ; Cellular: (209) 649-4967

9270 S Whitworth RD	Gustine	CA	95322
Mailing Address Number and Street	City	State	Zip Code

This operator is responsible for paying permit fees.

C. OWNERS

Lopes, Paul

Legal owner name: Lopes, Paul

Telephone no.: Landline: ; Cellular: (209) 649-4967

9270 S Whitworth RD	Gustine	CA	95322
Mailing Address Number and Street	City	State	Zip Code

This owner is responsible for paying permit fees.

AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	400	400	0	0
Number under roof	0	378	0	0	0	0
Maximum number	0	378	500	500	0	0
Average number	0	378	450	450	0	0
Avg live weight (lbs)	0	1,400	750	500		

Predominant milk cow breed: Holstein

Average milk production: 1 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 13,746.42 *tons per reporting period*

Total nitrogen from manure: 154,395.00 *lbs per reporting period*

After ammonia losses (30% loss applied): 108,076.50 *lbs per reporting period*

Total phosphorus from manure: 23,560.02 *lbs per reporting period*

Total potassium from manure: 1.00 *lbs per reporting period*

Total salt from manure: 86,921.10 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 17,115,360 *gallons*

Total nitrogen generated: 25,142.41 *lbs*

Total phosphorus generated: 6,713.30 *lbs*

Total potassium generated: 103,265.20 *lbs*

Total salt generated: 741,993.74 *lbs*

	17,115,360	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	17,115,360	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Ag Well #18	Ground water
Ag Well #20	Ground water
Ag Well #21	Ground water
Ag Well #23	Ground water
CCID Canal	Surface water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

Date	Material type / Description	Quantity	Reporting basis	Moist. (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/17/2023	Corral solids *Solid Manure from Tony Lopes Dairy*	2,120.00 *ton*	Dry-weight	59.1	13,900.00	3,400.00	21,600.00		0.00

No process wastewater nutrient imports entered.

Date	Material type / Description	Quantity	Reporting basis	Moisture (%)	N (%)	P (%)	K (%)	Salt (%)
06/25/2023	Solid commercial fertilizer *21-0-0*	116.50 *ton*	Dry-weight	0.1	21.000000	0.000000	0.000000	0.000000

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Commercial fertilizer / Other	48,881.07	0.00	0.00	0.00
Dry manure	24,104.82	5,896.14	37,457.86	0.00
Process wastewater	0.00	0.00	0.00	0.00
Total imports for all materials	72,985.89	5,896.14	37,457.86	0.00

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
Field 1	22	22	2	none	X070-X030-X020-XXXX
Field 18	123	123	2	manure	X070-X030-X014-XXXX
Field 19	90	90	2	manure	X070-X030-X014-XXXX
Field 2	86	86	0	none	X070-X010-X006-XXXX
Field 20	91	91	2	process wastewater	X070-X001-X010-XXXX
Field 21	84	84	2	manure	X070-X001-X009-XXXX X070-X001-X010-XXXX
Field 22	100	100	2	manure	X070-X001-X009-XXXX
Field 23	78	78	2	process wastewater	X070-X001-X009-XXXX
Field 24	87	87	1	none	X070-X001-X009-XXXX
Field 25	134	134	2	manure	X070-X001-X009-XXXX
Totals for areas that were used for application	700	700	14		
Totals for areas that were not used for application	195	195	3		
Land application area totals	895	895	17		

B. CROPS AND HARVESTS

Field 1

Field name: Field 1

10/27/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 22 Plant date: 10/27/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/18/2023	272.00 *ton*	As-is		61.8	4,970.00	1,400.00	19,900.00		8.74

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	14.00	140.00	22.40	116.20	0.00
Total actual harvest content	12.36	122.89	34.62	492.07	825.56

Field 1

05/27/2023: Corn, silage

Crop: Corn, silage Acres planted: 22 Plant date: 05/27/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/22/2023	468.00 *ton*	As-is		64.4	4,380.00	1,100.00	4,100.00		8.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	24.00	192.00	36.00	158.40	0.00
Total actual harvest content	21.27	186.35	46.80	174.44	1,348.01

Field 18

Field name: Field 18

10/19/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 123 Plant date: 10/19/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/20/2023	1,379.00 *ton*	As-is		50.4	4,510.00	900.00	5,800.00		8.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	14.00	140.00	22.40	116.20	0.00
Total actual harvest content	11.21	101.13	20.18	130.05	934.22

06/24/2023: Corn, silage

Crop: Corn, silage Acres planted: 123 Plant date: 06/24/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/17/2023	2,658.00 *ton*	As-is		64.4	4,380.00	1,050.00	4,090.00		8.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	24.00	192.00	36.00	158.40	0.00
Total actual harvest content	21.61	189.30	45.38	176.77	1,369.37

Field 19

Field name: Field 19

10/27/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 87 Plant date: 10/27/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/24/2023	1,304.00 *ton*	As-is		50.4	4,510.00	850.00	5,810.00		8.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	14.00	140.00	22.40	116.20	0.00
Total actual harvest content	14.99	135.20	25.48	174.17	1,248.96

05/26/2023: Corn, silage

Crop: Corn, silage Acres planted: 87 Plant date: 05/26/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/24/2023	1,999.00 *ton*	As-is		66.4	3,260.00	980.00	3,320.00		4.98

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	24.00	192.00	36.00	158.40	0.00
Total actual harvest content	22.98	149.81	45.03	152.57	768.94

Field 20

Field name: Field 20

10/28/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 91 Plant date: 10/28/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/24/2023	1,046.00 *ton*	As-is		68.8	3,010.00	480.00	3,280.00		9.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	14.00	140.00	22.40	116.20	0.00
Total actual harvest content	11.49	69.20	11.03	75.40	667.05

Field 20

05/15/2023: Corn, silage

Crop: Corn, silage Acres planted: 91 Plant date: 05/15/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/11/2023	1,623.00 *ton*	As-is		67.8	4,150.00	830.00	6,180.00		7.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	24.00	192.00	36.00	158.40	0.00
Total actual harvest content	17.84	148.03	29.61	220.44	884.41

Field 21

Field name: Field 21

10/25/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 84 Plant date: 10/25/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/21/2023	1,045.00 *ton*	As-is		47.0	6,050.00	1,110.00	8,190.00		9.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	14.00	140.00	22.40	116.20	0.00
Total actual harvest content	12.44	150.53	27.62	203.78	1,239.57

05/13/2023: Corn, silage

Crop: Corn, silage Acres planted: 84 Plant date: 05/13/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/12/2023	1,628.00 *ton*	As-is		55.2	4,970.00	970.00	6,560.00		7.28

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	24.00	192.00	36.00	158.40	0.00
Total actual harvest content	19.38	192.65	37.60	254.28	1,264.20

Field 22

Field name: Field 22

10/21/2022: Oats, hay

Crop: Oats, hay Acres planted: 100 Plant date: 10/21/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
06/10/2023	562.00 *ton*	As-is		10.0	22,400.00	2,820.00	20,500.00		7.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	3.00	120.00	19.50	99.00	0.00
Total actual harvest content	5.62	251.78	31.70	230.42	778.93

06/15/2023: Sudangrass, silage

Crop: Sudangrass, silage Acres planted: 100 Plant date: 06/15/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/30/2023	768.00 *ton*	As-is		73.6	11,100.00	710.00	10,800.00		14.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	176.00	27.20	192.00	0.00
Total actual harvest content	7.68	170.50	10.91	165.89	600.15

Field 23

Field name: Field 23

10/24/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 78 Plant date: 10/24/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/21/2023	903.00 *ton*	As-is		52.2	5,060.00	800.00	4,890.00		9.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	14.00	140.00	22.40	116.20	0.00
Total actual harvest content	11.58	117.16	18.52	113.22	1,018.21

Field 23

05/26/2023: Corn, silage

Crop: Corn, silage Acres planted: 78 Plant date: 05/26/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/22/2023	1,809.00 *ton*	As-is		67.2	2,850.00	690.00	2,740.00		3.71

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	24.00	192.00	36.00	158.40	0.00
Total actual harvest content	23.19	132.20	32.01	127.09	564.45

Field 24

Field name: Field 24

01/01/2023: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 87 Plant date: 01/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/01/2023	289.00 *ton*	As-is		10.0	31,500.00	2,300.00	23,800.00		8.79

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	7.00	420.00	37.80	294.00	0.00
Total actual harvest content	3.32	209.28	15.28	158.12	525.58

Field 25

Field name: Field 25

Field 25

10/30/2022: Oats, hay

Crop: Oats, hay Acres planted: 134 Plant date: 10/30/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
06/05/2023	626.00 *ton*	As-is		10.0	24,200.00	2,910.00	23,800.00		8.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	3.00	120.00	19.50	99.00	0.00
Total actual harvest content	4.67	226.11	27.19	222.37	681.13

06/11/2023: Sudangrass, silage

Crop: Sudangrass, silage Acres planted: 134 Plant date: 06/11/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/29/2023	1,304.00 *ton*	As-is		73.0	11,100.00	700.00	10,800.00		14.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	176.00	27.20	192.00	0.00
Total actual harvest content	9.73	216.04	13.62	210.20	777.73

NUTRIENT BUDGET

A. LAND APPLICATIONS

Field 1 - 10/27/2022: Oats, silage-soft dough

Field name: Field 1

Crop: Oats, silage-soft dough — Plant date: 10/27/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/15/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	0.00	0.00	0.00	142.87	2,430,000.00 *gal*
Application event totals		0.00	0.00	0.00	142.87	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/17/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	0.00	0.00	0.00	190.49	3,240,000.00 *gal*
Application event totals		0.00	0.00	0.00	190.49	

Field 1 - 05/27/2023: Corn, silage

Field name: Field 1

Crop: Corn, silage — Plant date: 05/27/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	0.00	0.00	0.00	222.24	3,780,000.00 *gal*
Application event totals		0.00	0.00	0.00	222.24	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/17/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
21-0-0	Solid commercial fertilizer	105.00	0.00	0.00	0.00	
Application event totals		105.00	0.00	0.00	0.00	

Field 1 - 05/27/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	0.00	0.00	0.00	190.49	3,240,000.00 *gal*
Application event totals		0.00	0.00	0.00	190.49	

07/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	0.00	0.00	0.00	190.49	3,240,000.00 *gal*
Application event totals		0.00	0.00	0.00	190.49	

07/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	0.00	0.00	0.00	158.74	2,700,000.00 *gal*
Application event totals		0.00	0.00	0.00	158.74	

08/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	0.00	0.00	0.00	142.87	2,430,000.00 *gal*
Application event totals		0.00	0.00	0.00	142.87	

08/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	0.00	0.00	0.00	142.87	2,430,000.00 *gal*
Application event totals		0.00	0.00	0.00	142.87	

09/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	0.00	0.00	0.00	127.00	2,160,000.00 *gal*
Application event totals		0.00	0.00	0.00	127.00	

Field 18 - 10/19/2022: Oats, silage-soft dough

Field 18 - 10/19/2022: Oats, silage-soft dough

Field name: Field 18

Crop: Oats, silage-soft dough

Plant date: 10/19/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/04/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	115.55	28.26	179.56	5,378.52	1,250.00 *ton*
Application event totals		115.55	28.26	179.56	5,378.52	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/20/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #18	Ground water	14.65	0.00	0.00	744.75	18,144,000.00 *gal*
Application event totals		14.65	0.00	0.00	744.75	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #18	Ground water	12.56	0.00	0.00	638.36	15,552,000.00 *gal*
Application event totals		12.56	0.00	0.00	638.36	

Field 18 - 06/24/2023: Corn, silage

Field name: Field 18

Crop: Corn, silage

Plant date: 06/24/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/17/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
21-0-0	Solid commercial fertilizer	105.00	0.00	0.00	0.00	
Application event totals		105.00	0.00	0.00	0.00	

Field 18 - 06/24/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/17/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	120.17	29.39	186.74	5,593.66	1,300.00 *ton*
Application event totals		120.17	29.39	186.74	5,593.66	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	0.00	0.00	0.00	238.50	22,680,000.00 *gal*
Application event totals		0.00	0.00	0.00	238.50	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	0.00	0.00	0.00	204.43	19,440,000.00 *gal*
Application event totals		0.00	0.00	0.00	204.43	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	0.00	0.00	0.00	170.36	16,200,000.00 *gal*
Application event totals		0.00	0.00	0.00	170.36	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	0.00	0.00	0.00	153.32	14,580,000.00 *gal*
Application event totals		0.00	0.00	0.00	153.32	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	0.00	0.00	0.00	170.36	16,200,000.00 *gal*
Application event totals		0.00	0.00	0.00	170.36	

Field 18 - 06/24/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	0.00	0.00	0.00	153.32	14,580,000.00 *gal*
Application event totals		0.00	0.00	0.00	153.32	

09/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	0.00	0.00	0.00	153.32	14,580,000.00 *gal*
Application event totals		0.00	0.00	0.00	153.32	

Field 19 - 10/27/2022: Oats, silage-soft dough

Field name: Field 19

Crop: Oats, silage-soft dough

Plant date: 10/27/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/05/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	150.83	42.04	245.33	0.00	500.00 *ton*
Application event totals		150.83	42.04	245.33	0.00	

11/27/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #18	Ground water	14.79	0.00	0.00	752.09	12,960,000.00 *gal*
Application event totals		14.79	0.00	0.00	752.09	

03/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #18	Ground water	11.83	0.00	0.00	601.67	10,368,000.00 *gal*
Application event totals		11.83	0.00	0.00	601.67	

Field 19 - 05/26/2023: Corn, silage

Field name: Field 19

Crop: Corn, silage

Plant date: 05/26/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/17/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
21-0-0	Solid commercial fertilizer	105.00	0.00	0.00	0.00	
Application event totals		105.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/17/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	107.17	26.21	166.53	4,988.30	820.00 *ton*
Application event totals		107.17	26.21	166.53	4,988.30	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	0.00	0.00	0.00	216.77	14,580,000.00 *gal*
Application event totals		0.00	0.00	0.00	216.77	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	0.00	0.00	0.00	192.68	12,960,000.00 *gal*
Application event totals		0.00	0.00	0.00	192.68	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	0.00	0.00	0.00	164.58	11,070,000.00 *gal*
Application event totals		0.00	0.00	0.00	164.58	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	0.00	0.00	0.00	164.58	11,070,000.00 *gal*
Application event totals		0.00	0.00	0.00	164.58	

Field 19 - 05/26/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	0.00	0.00	0.00	180.64	12,150,000.00 *gal*
Application event totals		0.00	0.00	0.00	180.64	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	0.00	0.00	0.00	168.60	11,340,000.00 *gal*
Application event totals		0.00	0.00	0.00	168.60	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	0.00	0.00	0.00	154.55	10,395,000.00 *gal*
Application event totals		0.00	0.00	0.00	154.55	

Field 20 - 10/28/2022: Oats, silage-soft dough

Field name: Field 20

Crop: Oats, silage-soft dough

Plant date: 10/28/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/24/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	68.05	20.29	185.43	2,697.17	6,808,320.00 *gal*
Ag Well #20	Ground water	8.95	0.00	0.00	562.54	8,640,000.00 *gal*
Application event totals		77.01	20.29	185.43	3,259.72	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #20	Ground water	10.45	0.00	0.00	656.30	10,080,000.00 *gal*
Application event totals		10.45	0.00	0.00	656.30	

Field 20 - 05/15/2023: Corn, silage

Field name: Field 20

Crop: Corn, silage

Plant date: 05/15/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #20	Ground water	14.92	0.00	0.00	937.57	14,400,000.00 *gal*
Application event totals		14.92	0.00	0.00	937.57	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #20	Ground water	13.43	0.00	0.00	843.82	12,960,000.00 *gal*
Application event totals		13.43	0.00	0.00	843.82	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #20	Ground water	12.19	0.00	0.00	765.69	11,760,000.00 *gal*
Application event totals		12.19	0.00	0.00	765.69	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	81.29	23.08	417.62	2,319.27	4,066,080.00 *gal*
Ag Well #20	Ground water	10.69	0.00	0.00	671.93	10,320,000.00 *gal*
Application event totals		91.98	23.08	417.62	2,991.20	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #20	Ground water	11.19	0.00	0.00	703.18	10,800,000.00 *gal*
Application event totals		11.19	0.00	0.00	703.18	

Field 20 - 05/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #20	Ground water	11.19	0.00	0.00	703.18	10,800,000.00 *gal*
Application event totals		11.19	0.00	0.00	703.18	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #20	Ground water	8.95	0.00	0.00	562.54	8,640,000.00 *gal*
Application event totals		8.95	0.00	0.00	562.54	

Field 21 - 10/25/2022: Oats, silage-soft dough

Field name: Field 21

Crop: Oats, silage-soft dough

Plant date: 10/25/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/05/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	140.60	39.19	228.68	0.00	450.00 *ton*
Application event totals		140.60	39.19	228.68	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/14/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #20	Ground water	9.70	0.00	0.00	609.42	8,640,000.00 *gal*
Application event totals		9.70	0.00	0.00	609.42	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #20	Ground water	8.08	0.00	0.00	507.85	7,200,000.00 *gal*
Application event totals		8.08	0.00	0.00	507.85	

Field 21 - 05/13/2023: Corn, silage

Field name: Field 21

Crop: Corn, silage

Plant date: 05/13/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #20	Ground water	12.12	0.00	0.00	761.78	10,800,000.00 *gal*
Application event totals		12.12	0.00	0.00	761.78	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/14/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	166.61	126.08	389.05	4,691.98	960.00 *ton*
Application event totals		166.61	126.08	389.05	4,691.98	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #20	Ground water	10.78	0.00	0.00	677.14	9,600,000.00 *gal*
Application event totals		10.78	0.00	0.00	677.14	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #20	Ground water	10.51	0.00	0.00	660.21	9,360,000.00 *gal*
Application event totals		10.51	0.00	0.00	660.21	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #20	Ground water	8.89	0.00	0.00	558.64	7,920,000.00 *gal*
Application event totals		8.89	0.00	0.00	558.64	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #20	Ground water	9.70	0.00	0.00	609.42	8,640,000.00 *gal*
Application event totals		9.70	0.00	0.00	609.42	

Field 21 - 05/13/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #20	Ground water	8.89	0.00	0.00	558.64	7,920,000.00 *gal*
Application event totals		8.89	0.00	0.00	558.64	

08/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #20	Ground water	8.08	0.00	0.00	507.85	7,200,000.00 *gal*
Application event totals		8.08	0.00	0.00	507.85	

Field 22 - 10/21/2022: Oats, hay

Field name: Field 22

Crop: Oats, hay

Plant date: 10/21/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/08/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #21	Ground water	12.39	0.00	0.00	687.21	13,500,000.00 *gal*
Application event totals		12.39	0.00	0.00	687.21	

11/06/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	131.22	36.57	213.44	0.00	500.00 *ton*
Application event totals		131.22	36.57	213.44	0.00	

11/09/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #21	Ground water	11.57	0.00	0.00	641.40	12,600,000.00 *gal*
Application event totals		11.57	0.00	0.00	641.40	

Field 22 - 10/21/2022: Oats, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #21	Ground water	12.39	0.00	0.00	687.21	13,500,000.00 *gal*
Application event totals		12.39	0.00	0.00	687.21	

Field 22 - 06/15/2023: Sudangrass, silage

Field name: Field 22

Crop: Sudangrass, silage

Plant date: 06/15/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #21	Ground water	12.39	0.00	0.00	687.21	13,500,000.00 *gal*
Application event totals		12.39	0.00	0.00	687.21	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/25/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
21-0-0	Solid commercial fertilizer	105.00	0.00	0.00	0.00	
Application event totals		105.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #21	Ground water	11.57	0.00	0.00	641.40	12,600,000.00 *gal*
Application event totals		11.57	0.00	0.00	641.40	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #21	Ground water	9.91	0.00	0.00	549.77	10,800,000.00 *gal*
Application event totals		9.91	0.00	0.00	549.77	

Field 22 - 06/15/2023: Sudangrass, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #21	Ground water	9.91	0.00	0.00	549.77	10,800,000.00 *gal*
Application event totals		9.91	0.00	0.00	549.77	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #21	Ground water	9.91	0.00	0.00	549.77	10,800,000.00 *gal*
Application event totals		9.91	0.00	0.00	549.77	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #21	Ground water	8.26	0.00	0.00	458.14	9,000,000.00 *gal*
Application event totals		8.26	0.00	0.00	458.14	

Field 23 - 10/24/2022: Oats, silage-soft dough

Field name: Field 23

Crop: Oats, silage-soft dough

Plant date: 10/24/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/30/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #23	Ground water	8.11	0.00	0.00	675.95	10,800,000.00 *gal*
Application event totals		8.11	0.00	0.00	675.95	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	95.18	20.44	348.42	2,150.00	3,971,520.00 *gal*
Ag Well #23	Ground water	4.73	0.00	0.00	394.30	6,300,000.00 *gal*
Application event totals		99.91	20.44	348.42	2,544.30	

Field 23 - 05/26/2023: Corn, silage

Field name: Field 23

Crop: Corn, silage

Plant date: 05/26/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #23	Ground water	9.46	0.00	0.00	788.60	12,600,000.00 *gal*
Application event totals		9.46	0.00	0.00	788.60	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #23	Ground water	7.44	0.00	0.00	619.62	9,900,000.00 *gal*
Application event totals		7.44	0.00	0.00	619.62	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #23	Ground water	6.08	0.00	0.00	506.96	8,100,000.00 *gal*
Application event totals		6.08	0.00	0.00	506.96	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #23	Ground water	8.11	0.00	0.00	675.95	10,800,000.00 *gal*
Application event totals		8.11	0.00	0.00	675.95	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #23	Ground water	6.76	0.00	0.00	563.29	9,000,000.00 *gal*
Application event totals		6.76	0.00	0.00	563.29	

Field 23 - 05/26/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	52.93	15.03	271.94	1,510.22	2,269,440.00 *gal*
Ag Well #23	Ground water	4.73	0.00	0.00	394.30	6,300,000.00 *gal*
Application event totals		57.66	15.03	271.94	1,904.52	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #23	Ground water	6.76	0.00	0.00	563.29	9,000,000.00 *gal*
Application event totals		6.76	0.00	0.00	563.29	

Field 24 - 01/01/2023: Alfalfa, hay

Field name: Field 24

Crop: Alfalfa, hay

Plant date: 01/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #23	Ground water	8.08	0.00	0.00	673.36	12,000,000.00 *gal*
Application event totals		8.08	0.00	0.00	673.36	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #23	Ground water	9.29	0.00	0.00	774.36	13,800,000.00 *gal*
Application event totals		9.29	0.00	0.00	774.36	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #23	Ground water	7.27	0.00	0.00	606.02	10,800,000.00 *gal*
Application event totals		7.27	0.00	0.00	606.02	

Field 24 - 01/01/2023: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #23	Ground water	7.27	0.00	0.00	606.02	10,800,000.00 *gal*
Application event totals		7.27	0.00	0.00	606.02	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #23	Ground water	7.27	0.00	0.00	606.02	10,800,000.00 *gal*
Application event totals		7.27	0.00	0.00	606.02	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #23	Ground water	6.06	0.00	0.00	505.02	9,000,000.00 *gal*
Application event totals		6.06	0.00	0.00	505.02	

Field 25 - 10/30/2022: Oats, hay

Field name: Field 25

Crop: Oats, hay

Plant date: 10/30/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/06/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	127.31	35.48	207.06	0.00	650.00 *ton*
Application event totals		127.31	35.48	207.06	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
12/01/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #21	Ground water	11.10	0.00	0.00	615.41	16,200,000.00 *gal*
Application event totals		11.10	0.00	0.00	615.41	

Field 25 - 10/30/2022: Oats, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #23	Ground water	6.03	0.00	0.00	502.76	13,800,000.00 *gal*
Application event totals		6.03	0.00	0.00	502.76	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #23	Ground water	7.87	0.00	0.00	655.77	18,000,000.00 *gal*
Application event totals		7.87	0.00	0.00	655.77	

Field 25 - 06/11/2023: Sudangrass, silage

Field name: Field 25

Crop: Sudangrass, silage

Plant date: 06/11/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #21	Ground water	13.56	0.00	0.00	752.17	19,800,000.00 *gal*
Application event totals		13.56	0.00	0.00	752.17	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/25/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
21-0-0	Solid commercial fertilizer	105.00	0.00	0.00	0.00	
Application event totals		105.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #21	Ground water	11.10	0.00	0.00	615.41	16,200,000.00 *gal*
Application event totals		11.10	0.00	0.00	615.41	

Field 25 - 06/11/2023: Sudangrass, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #21	Ground water	12.33	0.00	0.00	683.79	18,000,000.00 *gal*
Application event totals		12.33	0.00	0.00	683.79	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #21	Ground water	11.30	0.00	0.00	626.81	16,500,000.00 *gal*
Application event totals		11.30	0.00	0.00	626.81	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #21	Ground water	11.10	0.00	0.00	615.41	16,200,000.00 *gal*
Application event totals		11.10	0.00	0.00	615.41	

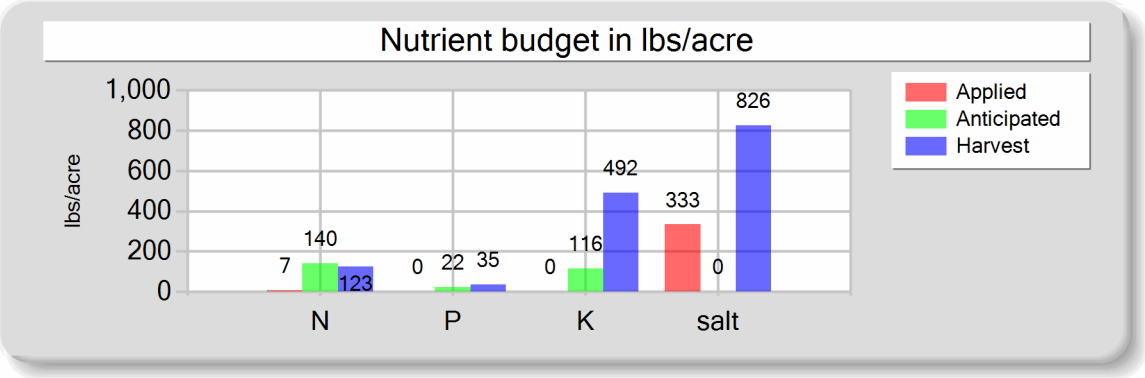
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #21	Ground water	12.33	0.00	0.00	683.79	18,000,000.00 *gal*
Application event totals		12.33	0.00	0.00	683.79	

B. NUTRIENT BUDGET

Field 1 - 10/27/2022: Oats, silage-soft dough

Field name: Field 1 Crop: Oats, silage-soft dough Plant date: 10/27/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	333.36
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	7.00	0.00	0.00	333.36
Anticipated crop nutrient removal	140.00	22.40	116.20	0.00
Actual crop nutrient removal	122.89	34.62	492.07	825.56
Nutrient balance	-115.89	-34.62	-492.07	-492.20
Applied to removed ratio	0.06	0.00	0.00	0.40

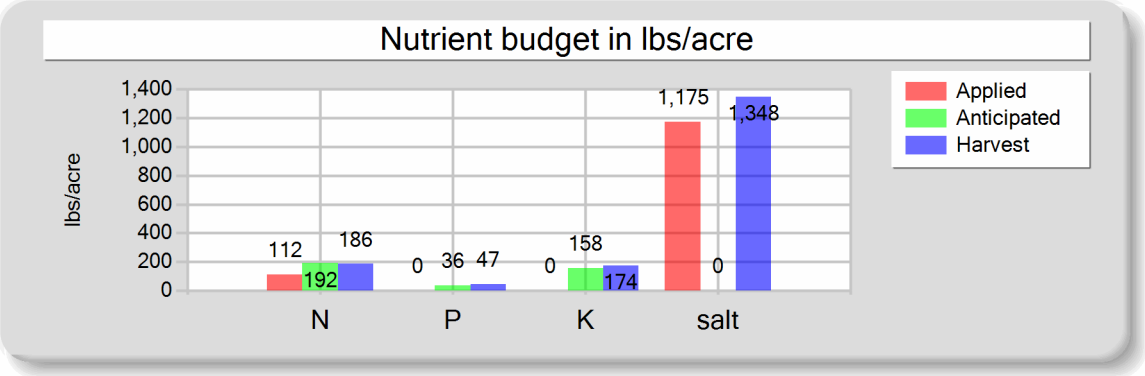
Fresh water applied	
5,670,000.00	*gallons*
208.81	*acre-inches*
9.49	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 1 - 05/27/2023: Corn, silage

Field name: Field 1 Crop: Corn, silage Plant date: 05/27/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	105.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	1,174.71
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	112.00	0.00	0.00	1,174.71
Anticipated crop nutrient removal	192.00	36.00	158.40	0.00
Actual crop nutrient removal	186.35	46.80	174.44	1,348.01
Nutrient balance	-74.35	-46.80	-174.44	-173.30
Applied to removed ratio	0.60	0.00	0.00	0.87

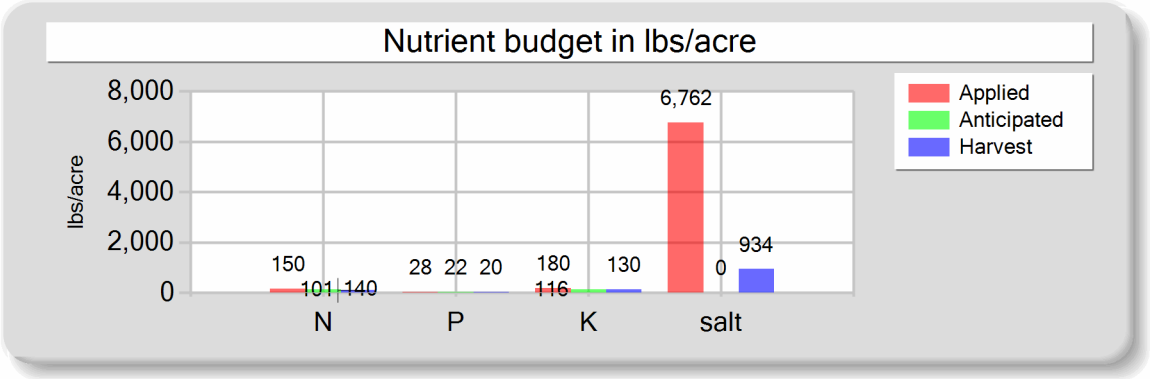
Fresh water applied	
19,980,000.00	*gallons*
735.80	*acre-inches*
33.45	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 18 - 10/19/2022: Oats, silage-soft dough

Field name: Field 18 Crop: Oats, silage-soft dough Plant date: 10/19/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	115.55	28.26	179.56	5,378.52
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	27.20	0.00	0.00	1,383.10
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	149.76	28.26	179.56	6,761.62
Anticipated crop nutrient removal	140.00	22.40	116.20	0.00
Actual crop nutrient removal	101.13	20.18	130.05	934.22
Nutrient balance	48.63	8.08	49.51	5,827.40
Applied to removed ratio	1.48	1.40	1.38	7.24

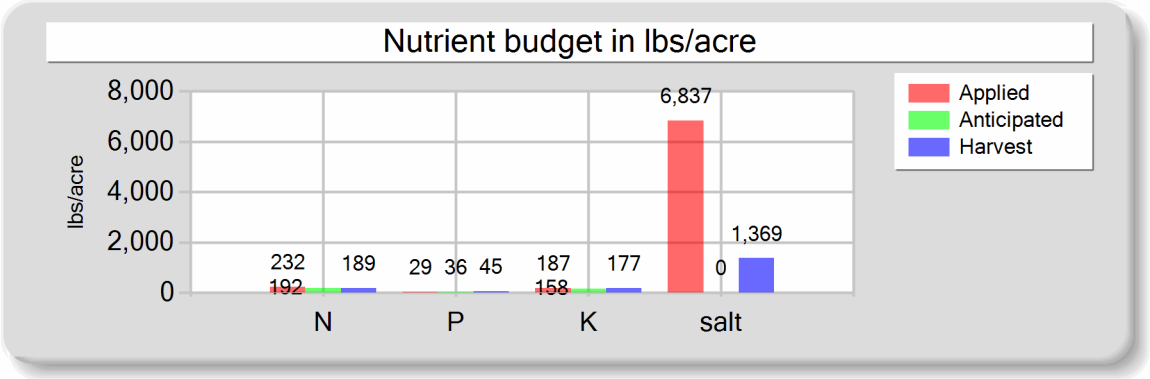
Fresh water applied	
33,696,000.00	*gallons*
1,240.91	*acre-inches*
10.09	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 18 - 06/24/2023: Corn, silage

Field name: Field 18 Crop: Corn, silage Plant date: 06/24/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	105.00	0.00	0.00	0.00
Dry manure	120.17	29.39	186.74	5,593.66
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	1,243.63
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	232.17	29.39	186.74	6,837.29
Anticipated crop nutrient removal	192.00	36.00	158.40	0.00
Actual crop nutrient removal	189.30	45.38	176.77	1,369.37
Nutrient balance	42.87	-15.99	9.98	5,467.92
Applied to removed ratio	1.23	0.65	1.06	4.99

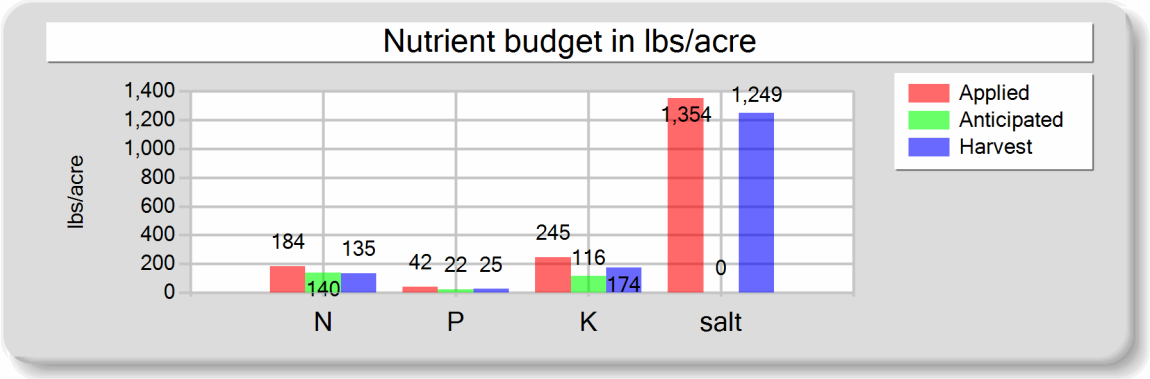
Fresh water applied	
118,260,000.00	*gallons*
4,355.11	*acre-inches*
35.41	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 19 - 10/27/2022: Oats, silage-soft dough

Field name: Field 19 Crop: Oats, silage-soft dough Plant date: 10/27/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	150.83	42.04	245.33	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	26.63	0.00	0.00	1,353.75
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	184.46	42.04	245.33	1,353.75
Anticipated crop nutrient removal	140.00	22.40	116.20	0.00
Actual crop nutrient removal	135.20	25.48	174.17	1,248.96
Nutrient balance	49.26	16.56	71.16	104.79
Applied to removed ratio	1.36	1.65	1.41	1.08

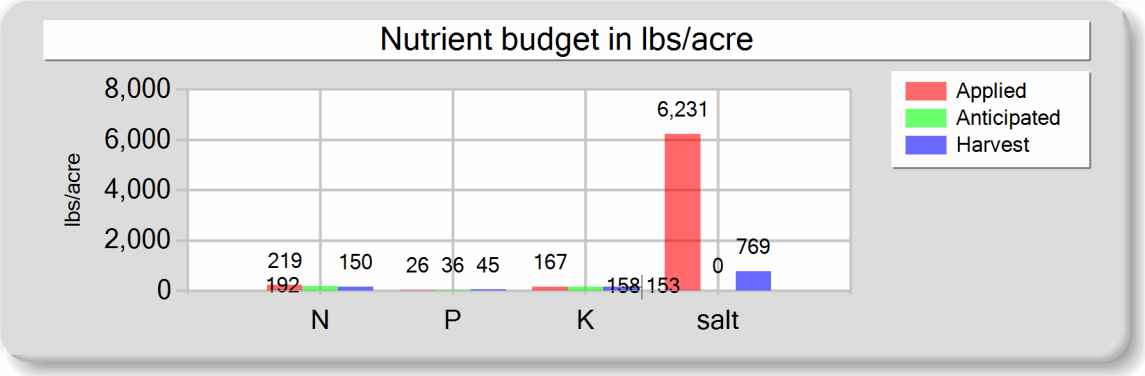
Fresh water applied	
23,328,000.00	*gallons*
859.09	*acre-inches*
9.87	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 19 - 05/26/2023: Corn, silage

Field name: Field 19 Crop: Corn, silage Plant date: 05/26/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	105.00	0.00	0.00	0.00
Dry manure	107.17	26.21	166.53	4,988.30
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	1,242.41
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	219.17	26.21	166.53	6,230.70
Anticipated crop nutrient removal	192.00	36.00	158.40	0.00
Actual crop nutrient removal	149.81	45.03	152.57	768.94
Nutrient balance	69.36	-18.82	13.97	5,461.76
Applied to removed ratio	1.46	0.58	1.09	8.10

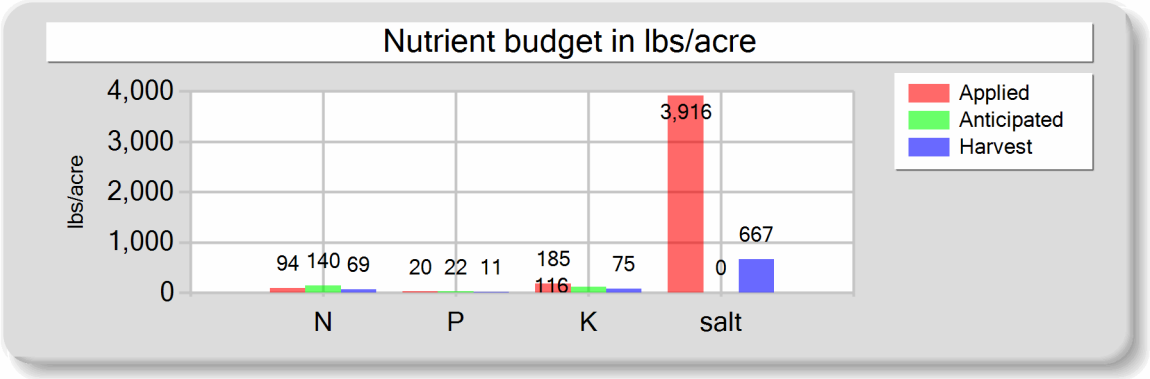
Fresh water applied	
83,565,000.00	*gallons*
3,077.41	*acre-inches*
35.37	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 20 - 10/28/2022: Oats, silage-soft dough

Field name: Field 20 Crop: Oats, silage-soft dough Plant date: 10/28/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	68.05	20.29	185.43	2,697.17
Fresh water	19.40	0.00	0.00	1,218.85
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	94.45	20.29	185.43	3,916.02
Anticipated crop nutrient removal	140.00	22.40	116.20	0.00
Actual crop nutrient removal	69.20	11.03	75.40	667.05
Nutrient balance	25.26	9.26	110.03	3,248.97
Applied to removed ratio	1.36	1.84	2.46	5.87

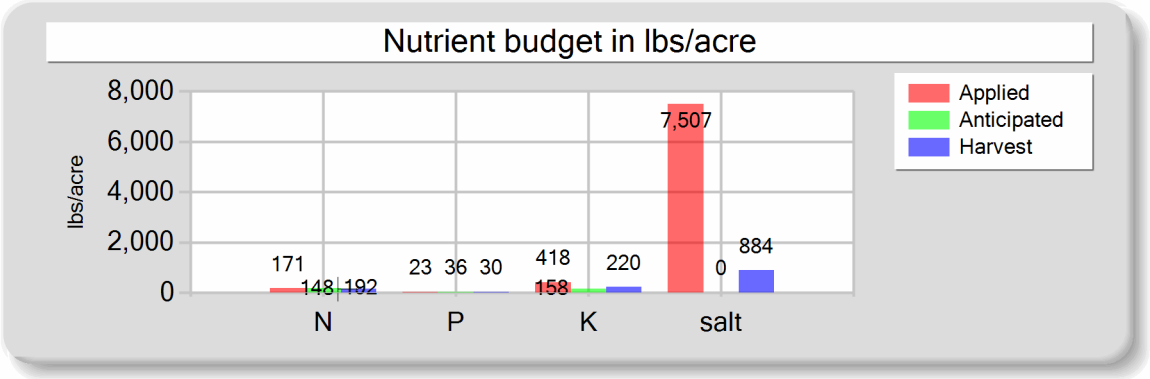
Fresh water applied	
18,720,000.00	*gallons*
689.39	*acre-inches*
7.58	*inches/acre*

Process wastewater applied	
6,808,320.00	*gallons*
250.73	*acre-inches*
2.76	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 20 - 05/15/2023: Corn, silage

Field name: Field 20 Crop: Corn, silage Plant date: 05/15/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	81.29	23.08	417.62	2,319.27
Fresh water	82.57	0.00	0.00	5,187.91
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	170.85	23.08	417.62	7,507.18
Anticipated crop nutrient removal	192.00	36.00	158.40	0.00
Actual crop nutrient removal	148.03	29.61	220.44	884.41
Nutrient balance	22.82	-6.53	197.18	6,622.77
Applied to removed ratio	1.15	0.78	1.89	8.49

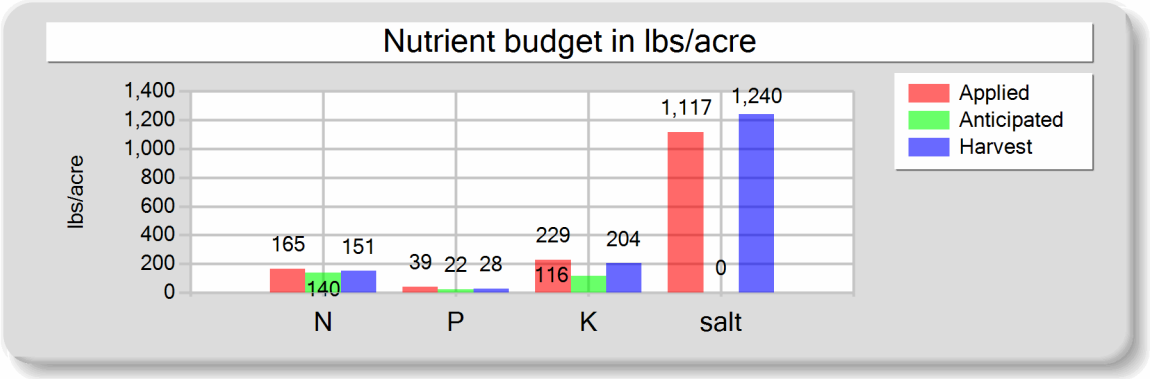
Fresh water applied	
79,680,000.00	*gallons*
2,934.34	*acre-inches*
32.25	*inches/acre*

Process wastewater applied	
4,066,080.00	*gallons*
149.74	*acre-inches*
1.65	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 21 - 10/25/2022: Oats, silage-soft dough

Field name: Field 21 Crop: Oats, silage-soft dough Plant date: 10/25/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	140.60	39.19	228.68	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	17.78	0.00	0.00	1,117.28
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	165.38	39.19	228.68	1,117.28
Anticipated crop nutrient removal	140.00	22.40	116.20	0.00
Actual crop nutrient removal	150.53	27.62	203.78	1,239.57
Nutrient balance	14.85	11.57	24.91	-122.29
Applied to removed ratio	1.10	1.42	1.12	0.90

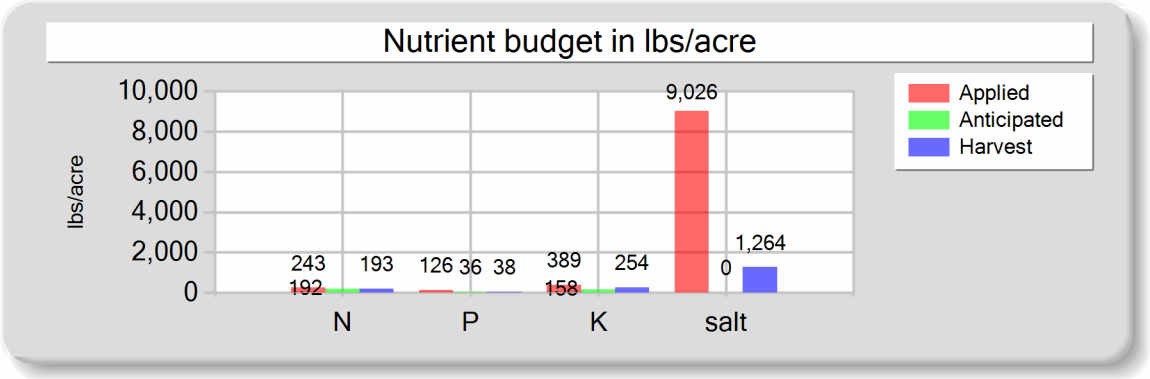
Fresh water applied	
15,840,000.00	*gallons*
583.33	*acre-inches*
6.94	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 21 - 05/13/2023: Corn, silage

Field name: Field 21 Crop: Corn, silage Plant date: 05/13/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	166.61	126.08	389.05	4,691.98
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	68.97	0.00	0.00	4,333.68
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	242.58	126.08	389.05	9,025.65
Anticipated crop nutrient removal	192.00	36.00	158.40	0.00
Actual crop nutrient removal	192.65	37.60	254.28	1,264.20
Nutrient balance	49.93	88.48	134.77	7,761.46
Applied to removed ratio	1.26	3.35	1.53	7.14

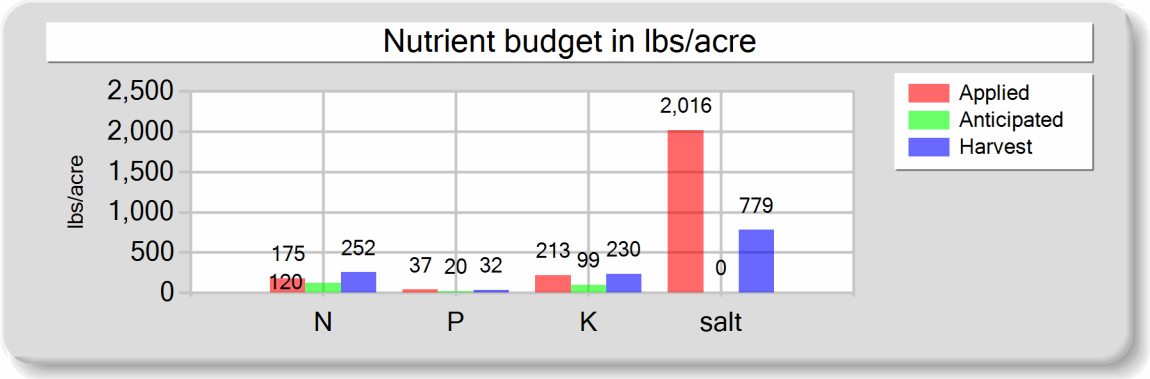
Fresh water applied	
61,440,000.00	*gallons*
2,262.63	*acre-inches*
26.94	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 22 - 10/21/2022: Oats, hay

Field name: Field 22 Crop: Oats, hay Plant date: 10/21/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	131.22	36.57	213.44	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	36.35	0.00	0.00	2,015.82
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	174.57	36.57	213.44	2,015.82
Anticipated crop nutrient removal	120.00	19.50	99.00	0.00
Actual crop nutrient removal	251.78	31.70	230.42	778.93
Nutrient balance	-77.20	4.88	-16.99	1,236.89
Applied to removed ratio	0.69	1.15	0.93	2.59

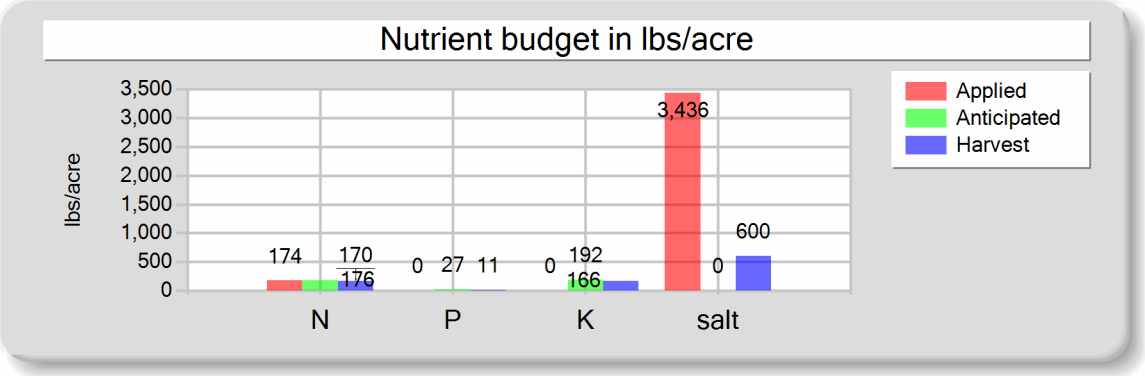
Fresh water applied	
39,600,000.00	*gallons*
1,458.33	*acre-inches*
14.58	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 22 - 06/15/2023: Sudangrass, silage

Field name: Field 22 Crop: Sudangrass, silage Plant date: 06/15/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	105.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	61.96	0.00	0.00	3,436.05
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	173.96	0.00	0.00	3,436.05
Anticipated crop nutrient removal	176.00	27.20	192.00	0.00
Actual crop nutrient removal	170.50	10.91	165.89	600.15
Nutrient balance	3.47	-10.91	-165.89	2,835.91
Applied to removed ratio	1.02	0.00	0.00	5.73

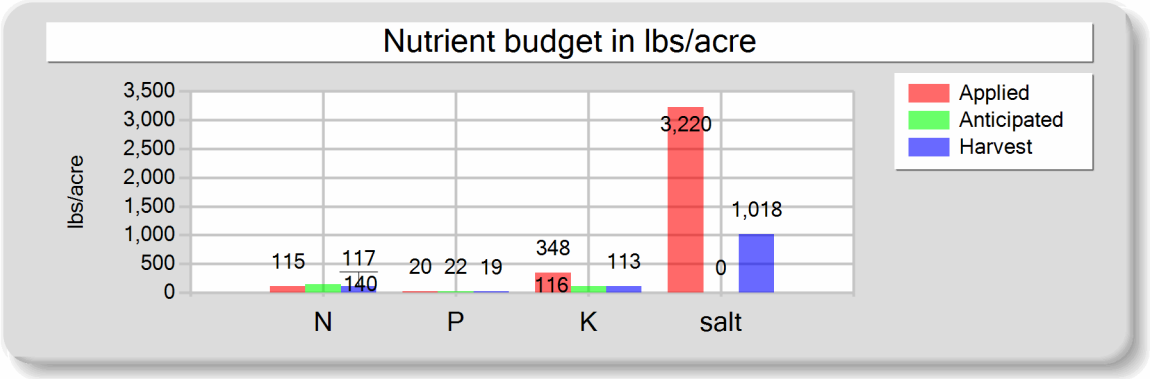
Fresh water applied	
67,500,000.00	*gallons*
2,485.80	*acre-inches*
24.86	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 23 - 10/24/2022: Oats, silage-soft dough

Field name: Field 23 Crop: Oats, silage-soft dough Plant date: 10/24/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	95.18	20.44	348.42	2,150.00
Fresh water	12.84	0.00	0.00	1,070.25
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	115.02	20.44	348.42	3,220.25
Anticipated crop nutrient removal	140.00	22.40	116.20	0.00
Actual crop nutrient removal	117.16	18.52	113.22	1,018.21
Nutrient balance	-2.14	1.91	235.20	2,202.04
Applied to removed ratio	0.98	1.10	3.08	3.16

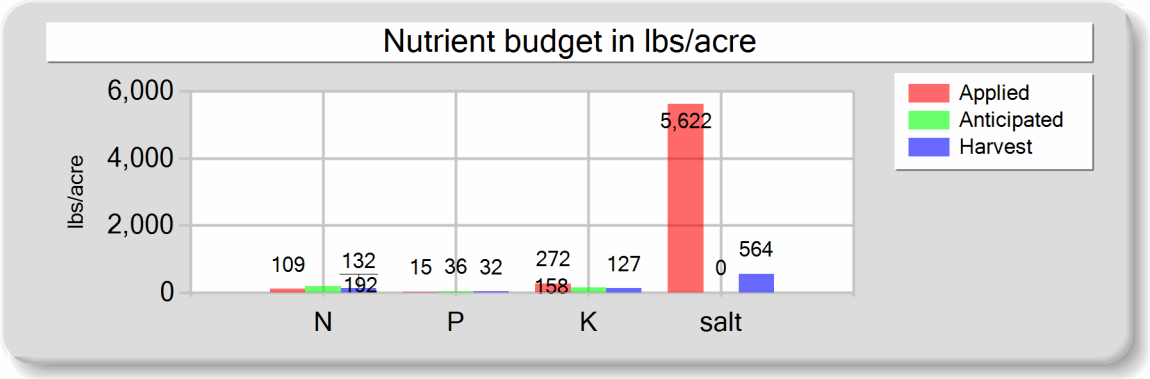
Fresh water applied	
17,100,000.00	*gallons*
629.73	*acre-inches*
8.07	*inches/acre*

Process wastewater applied	
3,971,520.00	*gallons*
146.26	*acre-inches*
1.88	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 23 - 05/26/2023: Corn, silage

Field name: Field 23 Crop: Corn, silage Plant date: 05/26/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	52.93	15.03	271.94	1,510.22
Fresh water	49.34	0.00	0.00	4,112.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	109.27	15.03	271.94	5,622.22
Anticipated crop nutrient removal	192.00	36.00	158.40	0.00
Actual crop nutrient removal	132.20	32.01	127.09	564.45
Nutrient balance	-22.92	-16.98	144.84	5,057.78
Applied to removed ratio	0.83	0.47	2.14	9.96

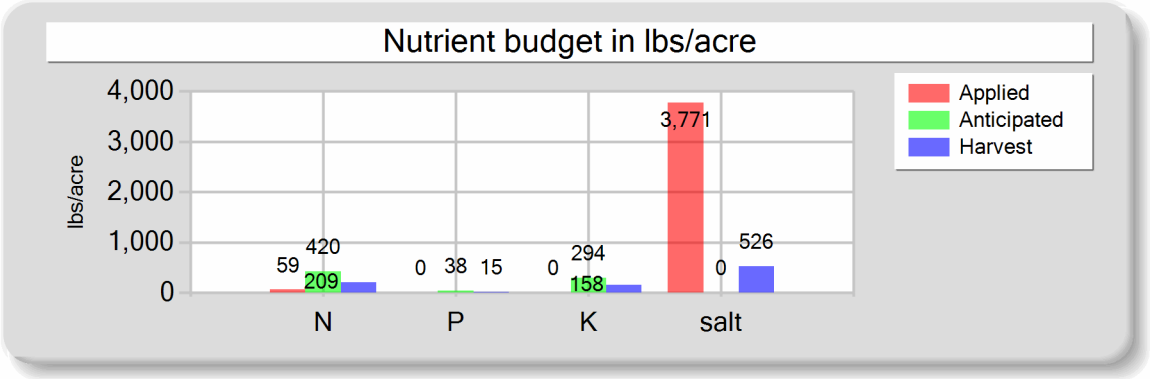
Fresh water applied	
65,700,000.00	*gallons*
2,419.51	*acre-inches*
31.02	*inches/acre*

Process wastewater applied	
2,269,440.00	*gallons*
83.58	*acre-inches*
1.07	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 24 - 01/01/2023: Alfalfa, hay

Field name: Field 24 Crop: Alfalfa, hay Plant date: 01/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	45.25	0.00	0.00	3,770.79
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	59.25	0.00	0.00	3,770.79
Anticipated crop nutrient removal	420.00	37.80	294.00	0.00
Actual crop nutrient removal	209.28	15.28	158.12	525.58
Nutrient balance	-150.03	-15.28	-158.12	3,245.21
Applied to removed ratio	0.28	0.00	0.00	7.17

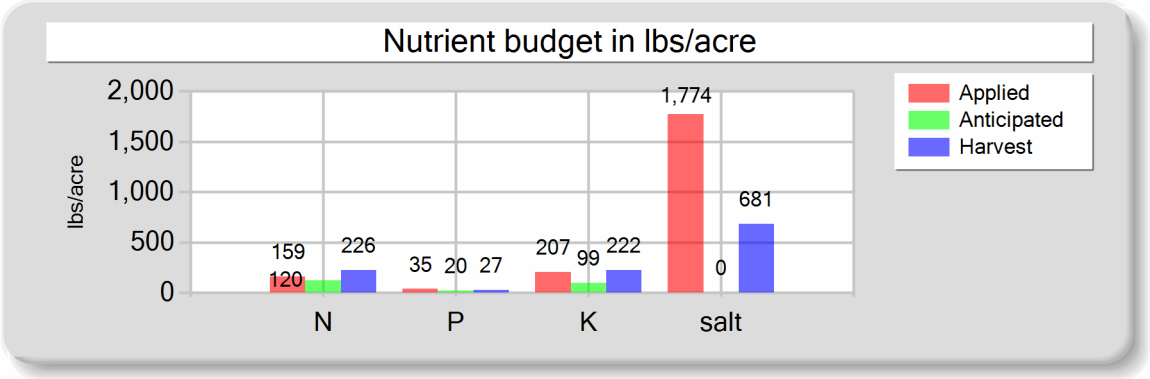
Fresh water applied	
67,200,000.00	*gallons*
2,474.75	*acre-inches*
28.45	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 25 - 10/30/2022: Oats, hay

Field name: Field 25 Crop: Oats, hay Plant date: 10/30/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	127.31	35.48	207.06	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	25.00	0.00	0.00	1,773.94
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	159.31	35.48	207.06	1,773.94
Anticipated crop nutrient removal	120.00	19.50	99.00	0.00
Actual crop nutrient removal	226.11	27.19	222.37	681.13
Nutrient balance	-66.80	8.29	-15.31	1,092.81
Applied to removed ratio	0.70	1.31	0.93	2.60

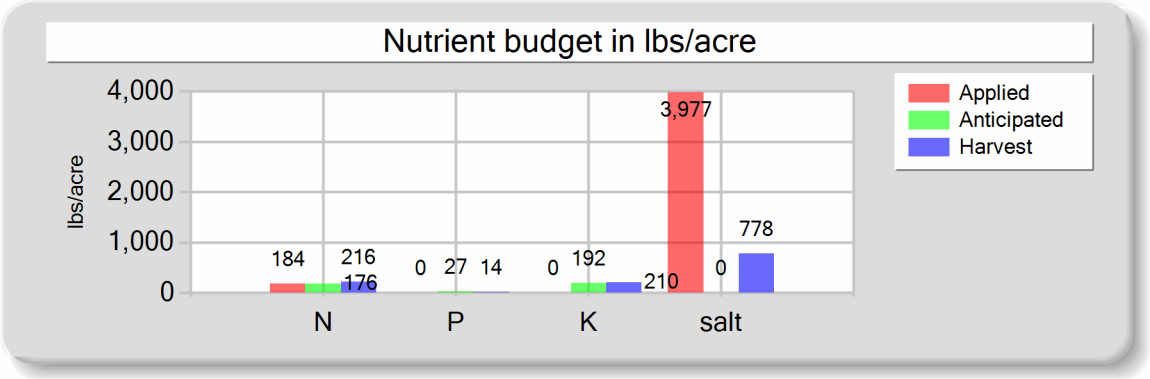
Fresh water applied	
48,000,000.00	*gallons*
1,767.68	*acre-inches*
13.19	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 25 - 06/11/2023: Sudangrass, silage

Field name: Field 25 Crop: Sudangrass, silage Plant date: 06/11/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	105.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	71.72	0.00	0.00	3,977.39
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	183.72	0.00	0.00	3,977.39
Anticipated crop nutrient removal	176.00	27.20	192.00	0.00
Actual crop nutrient removal	216.04	13.62	210.20	777.73
Nutrient balance	-32.31	-13.62	-210.20	3,199.66
Applied to removed ratio	0.85	0.00	0.00	5.11

Fresh water applied	
104,700,000.00	*gallons*
3,855.74	*acre-inches*
28.77	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Fall SM

Sample and source description: Fall SM

Sample date: 10/21/2022 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 47.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	24,900.00	6,940.00	40,500.00							0.00
DL	100.00	100.00	100.00							0.01

Fall SM

Sample and source description: Fall SM

Sample date: 10/21/2022 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 66.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	25,900.00	2,030.00	3,200.00							0.00
DL	100.00	100.00	100.00							0.01

Spring SM

Sample and source description: Spring SM

Sample date: 05/03/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 60.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	18,500.00	14,000.00	43,200.00	46,600.00	16,000.00	9,720.00	7,650.00	3,220.00		52.10
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	50.00		0.01

Spring SM Tony Lopes

Sample and source description: Spring SM Tony Lopes

Sample date: 05/03/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 59.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,900.00	3,400.00	21,600.00	22,700.00	17,500.00	5,510.00	4,950.00	16,100.00		64.70
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	50.00		0.01

Fall SM

Sample and source description: Fall SM

Sample date: 10/05/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 58.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	20,100.00	3,050.00	4,840.00							67.70
DL	100.00	100.00	100.00							0.01

B. PROCESS WASTEWATER ANALYSES

WWQ4

Sample and source description: WWQ4

Sample date: 12/21/2022 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	109.00	56.80	0.00	0.00	32.50	297.00								3,090.00	4,320
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWQ1

Sample and source description: WWQ1

Sample date: 03/10/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	224.00	137.00	0.00	0.00	48.10	820.00								5,430.00	5,060
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWQ3

Sample and source description: WWQ3

Sample date: 05/30/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	218.00	109.00	0.00	0.00	61.90	1,120.00								6,090.00	6,220
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWQ3

Sample and source description: WWQ3

Sample date: 10/02/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	344.00	216.00	0.00	0.00	51.00	1,200.00								7,450.00	5,420
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWQ4

Sample and source description: WWQ4

Sample date: 12/20/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	566.00	297.00	0.00	0.00	51.30	1,670.00	190.00	151.00	50.40	3,900.00	0.00	272.00	0.00	8,850.00	10,400
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	1.00	0.10	1.00	10

C. FRESH WATER ANALYSES

Ag Well #18

Ag Well #18

Sample description: Ag Well #18

Sample date: 01/03/2024 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	11.90	0.00	11.90								1,270.00	605
DL	0.50	0.50	0.50								1.00	10

Ag Well #20

Ag Well #20

Sample description: Ag Well #20

Sample date: 01/03/2024 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	11.30	0.00	8.50								1,540.00	710
DL	0.50	5.00	0.50								1.00	10

Ag Well #21

Ag Well #21

Sample description: Ag Well #21

Sample date: 01/03/2024 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	11.00	0.00	8.23								1,260.00	610
DL	0.50	0.50	0.50								1.00	10

Ag Well #23

Ag ell #23

Sample description: Ag ell #23

Sample date: 01/03/2024 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	7.02	0.00	4.22								1,240.00	585
DL	0.50	0.50	0.50								1.00	10

CCID Canal

CCID Canal

Sample description: CCID Canal

Sample date: 07/25/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								228.00	155
DL	1.00	0.50	0.50								10.00	10

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

Field 18 - 06/24/2023: Corn, silage

Field 18 - 06/24/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/23/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 64.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,380.00	1,050.00	4,090.00		8.90
DL	100.00	100.00	100.00		0.01

Field 19 - 10/27/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/03/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 50.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,510.00	850.00	5,810.00		8.40
DL	100.00	100.00	100.00		0.01

Field 19 - 05/26/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/26/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 66.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,260.00	980.00	3,320.00		4.98
DL	100.00	100.00	100.00		0.01

Field 20 - 10/28/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/03/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 68.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,010.00	480.00	3,280.00		9.30
DL	100.00	100.00	100.00		0.01

Field 20 - 05/15/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/26/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 67.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,150.00	830.00	6,180.00		7.70
DL	100.00	100.00	100.00		0.01

Field 21 - 10/25/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/03/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 47.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	6,050.00	1,110.00	8,190.00		9.40
DL	100.00	100.00	100.00		0.01

Field 21 - 05/13/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/26/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 55.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,970.00	970.00	6,560.00		7.28
DL	100.00	100.00	100.00		0.01

Field 22 - 10/21/2022: Oats, hay

Oat Hay

Sample and source description: Oat Hay

Sample date: 05/03/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 10.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	22,400.00	2,820.00	20,500.00		7.70
DL	100.00	100.00	100.00		0.01

Field 22 - 06/15/2023: Sudangrass, silage

Sudan

Sample and source description: Sudan

Sample date: 09/04/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 73.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,100.00	710.00	10,800.00		14.80
DL	100.00	100.00	100.00		0.01

Field 23 - 10/24/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/03/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 52.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,060.00	800.00	4,890.00		9.20
DL	100.00	100.00	100.00		0.01

Field 23 - 05/26/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/26/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 67.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	2,850.00	690.00	2,740.00		3.71
DL	100.00	100.00	100.00		0.01

Field 24 - 01/01/2023: Alfalfa, hay

Alfalfa

Sample and source description: Alfalfa

Sample date: 06/16/2022 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 10.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	31,500.00	2,300.00	23,800.00		8.79
DL	100.00	100.00	100.00		0.01

Field 25 - 10/30/2022: Oats, hay

Oat Hay

Sample and source description: Oat Hay

Sample date: 05/03/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 10.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	24,200.00	2,910.00	23,800.00		8.10
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

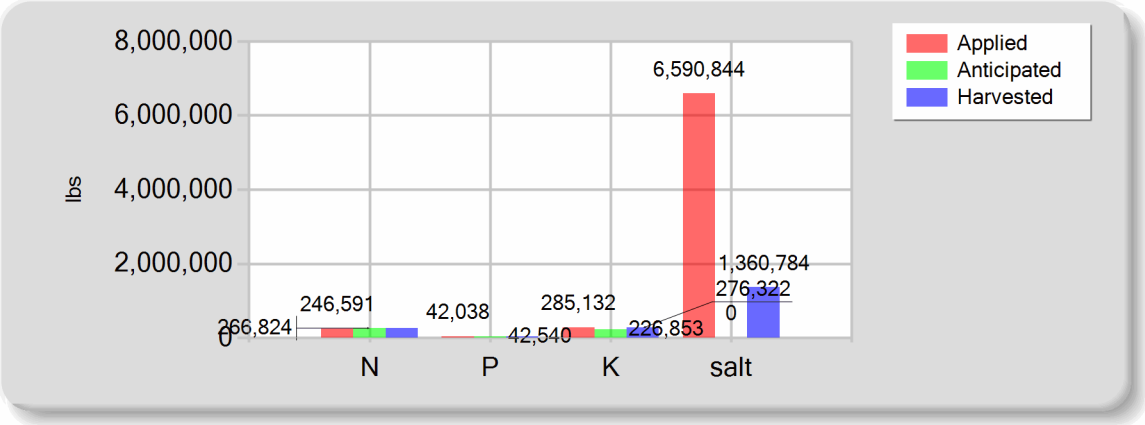
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

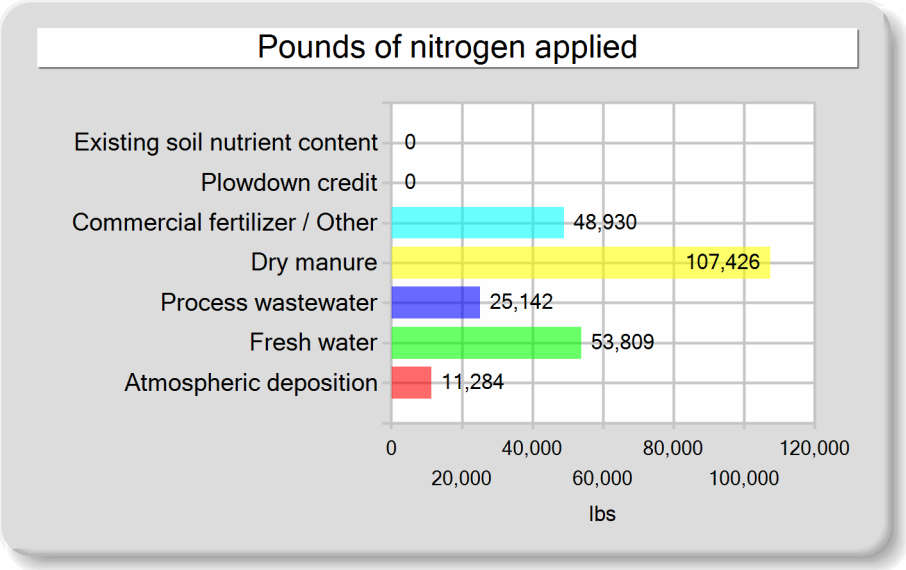
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

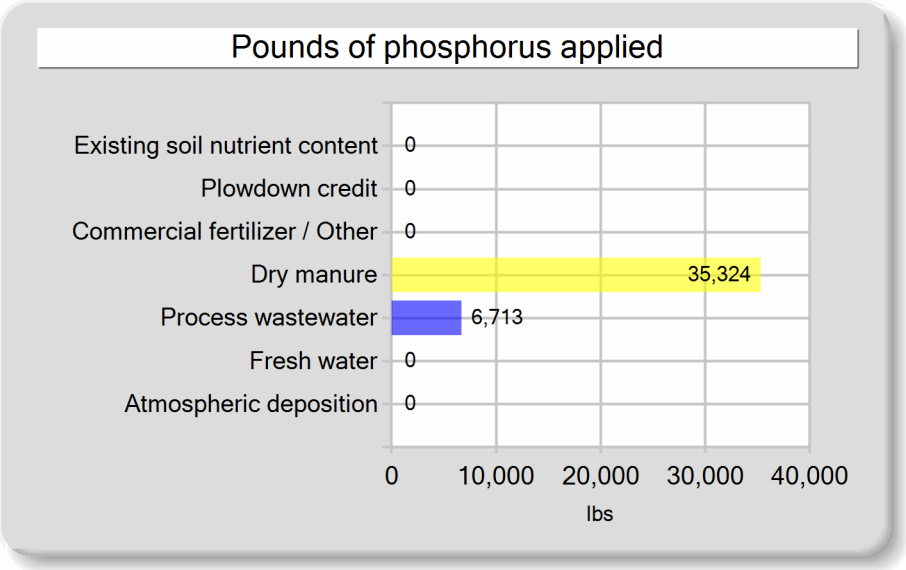
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	48,930.00	0.00	0.00	0.00
Dry manure	107,426.11	35,324.36	181,866.49	2,177,685.10
Process wastewater	25,142.41	6,713.30	103,265.20	741,993.74
Fresh water	53,808.61	0.00	0.00	3,671,165.32
Atmospheric deposition	11,284.00	0.00	0.00	0.00
Total nutrients applied	246,591.13	42,037.66	285,131.69	6,590,844.16
Anticipated crop nutrient removal	266,824.00	42,540.40	226,853.00	0.00
Actual crop nutrient removal	255,047.28	40,166.38	276,322.24	1,360,783.85
Nutrient balance	-8,456.15	1,871.28	8,809.45	5,230,060.31
Applied to removed ratio	0.97	1.05	1.03	4.84

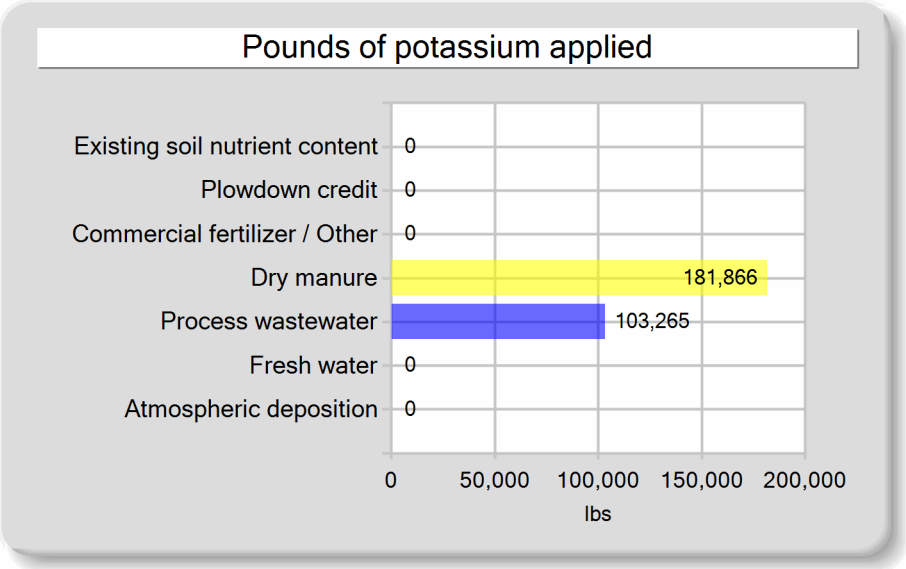
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

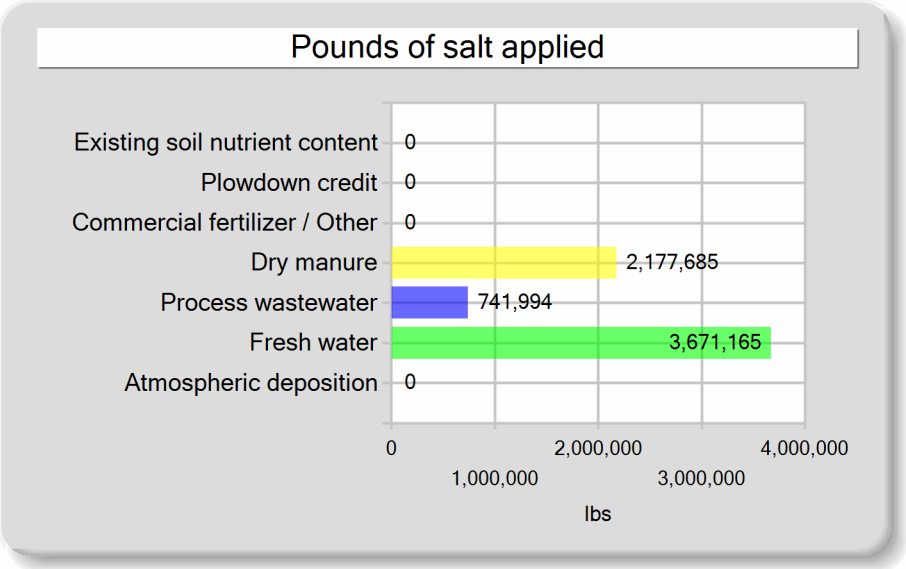


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	No
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Paul Lopes
Paul Lopes (Apr 23, 2024 14:17 PDT)

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Paul Lopes	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
Apr 23, 2024	
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

www.denelelabs.com

Compliance Analysis Report

Customer
P & D Dairy
9270 Whitworth Rd
Gustine, CA 95322

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER: North
RECEIVED DATE: 1/3/24 9:10 AM
SUBMITTED BY: Joe Ramos
ANALYZED DATE: 1/5/24

Irrigation / Ground Water (H7)
SOURCE: Irrigation Water

NOTES: Temperature was above qualifying at 10.8c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Ag Well 18	T4003002A	Electrical Conductivity (EC)	EPA 120.1 and	1.27	mmhos/cm
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Nitrate Nitrogen (NO3-N)	EPA 353.2	9.12	mg/L
		Total Dissolved Solids (TDS)	SM 2540 C-1997 and	605	mg/L
		Total Nitrogen (TN)	SM 4500-NH3 D,C	11.9	mg/L
		Soluble Salts (SALT-SOL)	Calculation	813	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	Sample mg/L	Dup mg/L	RPD %
Ammonia Nitrogen (NH3-N)	96.0	95.8 (T4003010D)	94.0 (T4003010D)			1.89
Electrical Conductivity (EC)	96.9			1.49 (T3362090A)	1.55 (Dup)	3.69
Nitrate Nitrogen (NO3-N)	103	102 (T3362098C)	101 (T3362098C)			0.866
Total Dissolved Solids (TDS)	103			435 (T4003007F)	440 (Dup)	1.14
Total Nitrogen (TN)	101			5.60 (T3362069C)	5.60 (null)	0.000

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ELAP Certificate No.: 2714

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Compliance Analysis Report

Customer
P & D Dairy
9270 Whitworth Rd
Gustine, CA 95322

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER: North
RECEIVED DATE: 1/3/24 9:10 AM
SUBMITTED BY: Joe Ramos
ANALYZED DATE: 1/9/24

Irrigation / Ground Water (H7)
SOURCE: Irrigation Water

NOTES: Temperature was above qualifying at 11.2c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Ag Well 20	T4003002B	Electrical Conductivity (EC)	EPA 120.1 and	1.54	mmhos/cm
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Nitrate Nitrogen (NO3-N)	EPA 353.2	8.50	mg/L
		Total Dissolved Solids (TDS)	SM 2540 C-1997 and	710	mg/L
		Total Nitrogen (TN)	SM 4500-NH3 D,C	11.3	mg/L
		Soluble Salts (SALT-SOL)	Calculation	984	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	Sample mg/L	Dup mg/L	RPD %
Ammonia Nitrogen (NH3-N)	100	97.0 (T3362098B)	95.2 (T3362098B)			1.87
Electrical Conductivity (EC)	99.2			1.54 (T4003002B)	1.52 (Dup)	1.05
Nitrate Nitrogen (NO3-N)	103	102 (T3362098C)	101 (T3362098C)			0.866
Total Dissolved Solids (TDS)	103			435 (T4003007F)	440 (Dup)	1.14
Total Nitrogen (TN)	101			5.60 (T3362069C)	5.60 (null)	0.000

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Compliance Analysis Report

Customer
P & D Dairy
9270 Whitworth Rd
Gustine, CA 95322

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER: North
RECEIVED DATE: 1/3/24 9:10 AM
SUBMITTED BY: Joe Ramos
ANALYZED DATE: 1/9/24

Irrigation / Ground Water (H7)
SOURCE: Irrigation Water

NOTES: Temperature was above qualifying at 11.2c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Ag Well 21	T4003002C	Electrical Conductivity (EC)	EPA 120.1 and	1.26	mmhos/cm
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Nitrate Nitrogen (NO3-N)	EPA 353.2	8.23	mg/L
		Total Dissolved Solids (TDS)	SM 2540 C-1997 and	610	mg/L
		Total Nitrogen (TN)	SM 4500-NH3 D,C	11.0	mg/L
		Soluble Salts (SALT-SOL)	Calculation	806	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	Sample mg/L	Dup mg/L	RPD %
Ammonia Nitrogen (NH3-N)	100	97.0 (T3362098B)	95.2 (T3362098B)			1.87
Electrical Conductivity (EC)	99.2			1.54 (T4003002B)	1.52 (Dup)	1.05
Nitrate Nitrogen (NO3-N)	103	102 (T3362098C)	101 (T3362098C)			0.866
Total Dissolved Solids (TDS)	103			435 (T4003007F)	440 (Dup)	1.14
Total Nitrogen (TN)	101			5.60 (T3362069C)	5.60 (null)	0.000

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Compliance Analysis Report

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P & D Dairy
9270 Whitworth Rd
Gustine, CA 95322

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER: North
RECEIVED DATE: 1/3/24 9:10 AM
SUBMITTED BY: Joe Ramos
ANALYZED DATE: 1/9/24

Irrigation / Ground Water (H7)
SOURCE: Irrigation Water

NOTES: Temperature was above qualifying at 11.2c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Ag Well 23	T4003002D	Electrical Conductivity (EC)	EPA 120.1 and	1.24	mmhos/cm
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Nitrate Nitrogen (NO3-N)	EPA 353.2	4.22	mg/L
		Total Dissolved Solids (TDS)	SM 2540 C-1997 and	585	mg/L
		Total Nitrogen (TN)	SM 4500-NH3 D,C	7.02	mg/L
		Soluble Salts (SALT-SOL)	Calculation	795	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	Sample mg/L	Dup mg/L	RPD %
Ammonia Nitrogen (NH3-N)	100	97.0 (T3362098B)	95.2 (T3362098B)			1.87
Electrical Conductivity (EC)	99.2			1.54 (T4003002B)	1.52 (Dup)	1.05
Nitrate Nitrogen (NO3-N)	103	102 (T3362098C)	101 (T3362098C)			0.866
Total Dissolved Solids (TDS)	103			435 (T4003007F)	440 (Dup)	1.14
Total Nitrogen (TN)	101			5.60 (T3362069C)	5.60 (null)	0.000

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

www.denelelabs.com

Compliance Analysis Report

Customer
P & D Dairy
9270 Whitworth Rd
Gustine, CA 95322

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER: North
RECEIVED DATE: 1/3/24 9:10 AM
SUBMITTED BY: Joe Ramos
ANALYZED DATE: 1/9/24

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 11.2c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW 1	T4003002E	Electrical Conductivity (EC)	EPA 120.1 and	1.30	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	16.6	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	829	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	100	97.0 (T3362098B)	95.2 (T3362098B)	1.87
Electrical Conductivity (EC)	99.2			1.05
Nitrate Nitrogen (NO3-N)	103	102 (T3362098C)	101 (T3362098C)	0.866

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Grower

Consultant:
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2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER: North
RECEIVED DATE: 1/3/24 9:10 AM
SUBMITTED BY: Joe Ramos
ANALYZED DATE: 1/9/24

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 11.4c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW 2	T4003002F	Electrical Conductivity (EC)	EPA 120.1 and	1.31	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	17.9	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	836	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	100	97.0 (T3362098B)	95.2 (T3362098B)	1.87
Electrical Conductivity (EC)	99.2			1.05
Nitrate Nitrogen (NO3-N)	103	102 (T3362098C)	101 (T3362098C)	0.866

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Compliance Analysis Report

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P & D Dairy
9270 Whitworth Rd
Gustine, CA 95322

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Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER: North
RECEIVED DATE: 1/3/24 9:10 AM
SUBMITTED BY: Joe Ramos
ANALYZED DATE: 1/9/24

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 12.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW 3	T4003002G	Electrical Conductivity (EC)	EPA 120.1 and	1.34	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	17.3	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	857	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	100	97.0 (T3362098B)	95.2 (T3362098B)	1.87
Electrical Conductivity (EC)	99.2			1.05
Nitrate Nitrogen (NO3-N)	103	102 (T3362098C)	101 (T3362098C)	0.866

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ELAP Certificate No.: 2714

Denele Analytical, Inc.

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Phone (209) 634-9055 - Fax (209) 634-9057

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Compliance Analysis Report

Customer
P & D Dairy
9270 Whitworth Rd
Gustine, CA 95322

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER: North
RECEIVED DATE: 1/3/24 9:10 AM
SUBMITTED BY: Joe Ramos
ANALYZED DATE: 1/9/24

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 12.4c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW Trailers	T4003002H	Electrical Conductivity (EC)	EPA 120.1 and	1.32	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	6.39	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	847	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	100	97.0 (T3362098B)	95.2 (T3362098B)	1.87
Electrical Conductivity (EC)	99.2			1.05
Nitrate Nitrogen (NO3-N)	103	102 (T3362098C)	101 (T3362098C)	0.866

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ELAP Certificate No.: 2714

DENELE ANALYTICAL, INC.

865 S. KILROY
TURLOCK, CA 95380

PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Client: P & D Dairy North
Mailing address:
(not facility address)
Phone#:

Authorized Copy Released To: F+R
Sampled By:
Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
- H1 EC*, NO3-N, NH4-N*
- H6 EC, TN, TDS
- H7 Combination of H1 & H6
- H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
- H3 EC*, NH4-N, TKN, TP, K, TDS
- H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
- H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
- H8 Combination of H3 &H5
- H9 Combination of H4 &H5

TILE DRAIN
- H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*

___ (Check here if performed)

DISCHARGE WATER
- E1 EC*, Temp*, pH*, NO3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
- E2 EC,* Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
- E3 EC*, Temp*, pH*, NO3-N, NH4-N*, TP, Total & Fecal Coliforms
- E4 EC*, Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
- M1 % Moisture
- M3 TN TP, K, % Moisture
- M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
- M7 Combination of M3 & M4

PLANT TISSUE
- F18 % Moisture, Ash, TN, P, K, DM and as received
- F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
- F19+ % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
- K3 Soluble Phosphorus
- K4 1ft NO3, Organic Matter
- K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
- K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
- K7 Combination of K3 & K4
- K8 Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	H1	H3	M3	F18	OTHER
1	1/2	11:30	Ag Well 18		10.8°c						H7
2	1/2	11.40	Ag well 20		11.2°c						H7
3	1/2	11.45	Ag well 21		11.2°						H7
4	1/2	11.50	Ag well 23		11.2°						H7
5	1/2	12:00	DW 1		11.2°		X				
6	1/2	12:05	DW 2		11.4°		X				
7	1/2	12:10	DW 3		12.0°		X				
8	1/2	12:20	DW Trailers		12.4°c		X				
9											
10											

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	Logged in By:	LABORATORY NUMBER
1st						
2nd		DENELE ANALYTICAL, INC.	1-3-24 9:10			T4003002

(LABORATORY USE ONLY)

Required to be submitted to any agency? (YES / NO)

P&D Dairy North 2023 AR Complete

Final Audit Report 2024-04-23

Created:	2024-04-23
By:	Joe Ramos (jramos@fragservices.com)
Status:	Signed
Transaction ID:	CBJCHBCAABAAtAg1X9SsjsqRVjHqV5nSyTeVSElwJxFU

"P&D Dairy North 2023 AR Complete" History

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Signer pandd95@gmail.com entered name at signing as Paul Lopes
2024-04-23 - 9:17:41 PM GMT

Document e-signed by Paul Lopes (pandd95@gmail.com)
Signature Date: 2024-04-23 - 9:17:43 PM GMT - Time Source: server

Agreement completed.
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